

Multiscale Statistical Connectomics Reveal a Scale-Invariant Functional Connectivity Phenotyping in Glioblastoma

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Abstract

We apply statistical connectomics methods to characterize functional connectivity (FC) alterations in patients with glioblastoma (GBM). Using omnibus spectral embedding of resting-state fMRI data from 217 GBM patients and 252 age-matched healthy reference subjects, we discover that the networks consistently have an 8-dimensional latent structure across all functional parcellation granularities. Further multidimensional scaling recovers a low-dimensional representation in which patients separate from reference subjects with large effect size (Cohen’s $d \approx 1.2$). Remarkably, this separation is also *scale-invariant*: subject-level “patient-likeness” scores correlate at $r > 0.95$ across parcellation resolutions from 100 to 600 parcels. We introduce a continuous projection score that captures degree of FC disruption along the patient-reference axis. This score correlates more strongly with individual FC edges than binary diagnostic labels (21% increase in mean $|r|$; Wilcoxon $p < 10^{-10}$), demonstrating that the spectral embedding extracts graded FC variation beyond categorical group membership. Characterizing the FC signature of “patient-likeness,” we find that high-scoring individuals—whether patients or reference subjects—exhibit reduced connectivity in somatomotor and visual networks, with the strongest effects in inter-hemispheric sensorimotor connections ($\Delta r \approx -0.17$). The same edges define patient-likeness in both groups ($r = 0.63$ between within-group difference patterns), suggesting the projection captures a continuous FC phenotype that exists as normal variation in healthy individuals and is exaggerated in GBM. These findings support the conceptualization of GBM as a network-level disease and demonstrate the utility of spectral connectomics for discovering interpretable, low-dimensional biomarkers of distributed FC disruption.

1 Introduction

1.1 Glioblastoma as a Network Disease

Glioblastoma (GBM) is the most common and aggressive primary malignant brain tumor, with a median overall survival of less than two years despite multimodal treatment (Stupp et al., 2005; Weller et al., 2014). GBM is characterized by major metabolic reprogramming that sustains growth and modifies the tumor microenvironment (Agnihotri and Zadeh, 2016; Alzoughbi and et al., 2014; Dong and et al., 2019; Vander Heiden et al., 2009). These metabolic changes are not confined to the lesion itself: cerebrospinal fluid (CSF) metabolomic studies in malignant and high-grade gliomas show broad differences in metabolite profiles between patients and non-malignant controls and link specific CSF metabolites to tumor grade and prognosis (Locasale and et al., 2012; Nakamizo et al., 2013). Supporting a brain-wide disease view, resting-state functional magnetic resonance imaging (rs-fMRI) studies in GBM report global alterations in intrinsic activity and functional connectivity extending beyond the visible lesion (Daniel et al., 2021b; Park et al., 2023b).

In high-grade glioma, preoperative resting-state functional connectivity (FC) and tumor-network relationships predict postoperative functional status (KPS) (Luckett et al., 2024). In GBM, FC between tumor regions and large-scale resting-state networks and whole-brain spectral properties of the rs-fMRI signal (e.g., $1/f$ power-law exponents and fALFF) are associated with overall survival (Daniel et al., 2021a; Luckett et al., 2023; Park et al., 2023b). These observations suggest that distributed rs-fMRI metrics relate to clinically meaningful properties of the tumor-bearing brain, motivating quantitative frameworks to describe such distributed alterations.

1.2 Resting-State Functional Connectivity (rsFC)

Resting-state functional MRI (rs-fMRI) measures spontaneous fluctuations in the blood-oxygen-level-dependent (BOLD) signal while subjects rest without performing explicit tasks (Biswal et al., 1995; Fox and Raichle, 2007). Temporally correlated BOLD fluctuations between brain regions define functional connectivity, which in turn is used to characterize the hierarchical intrinsic functional architecture of the brain (Yeo et al., 2011; Power et al., 2011).

Rs-fMRI is particularly valuable in brain tumor patients as it requires no task performance and can be acquired alongside clinical imaging (Hacker et al., 2019; Park et al., 2023b). Canonical resting-state networks including default mode, frontoparietal control, dorsal and ventral attention, somatomotor, visual, and limbic systems are robustly reproducible in humans (Yeo et al., 2011), and their disruption in GBM has been linked to clinical symptoms and outcome (Daniel et al., 2021a,b; Luckett et al., 2023).

1.3 The Challenge of Multiscale Analysis

Constructing FC networks requires parcellating the cortex into functionally discrete regions, and both the choice of atlas and parcellation granularity materially affect downstream results (Arslan et al., 2018; Park et al., 2023a). Building on prior work in glioma showing that functionally derived atlases improve patient-reference separation (Park et al., 2023a), we employ the Schaefer local-global parcellation (Schaefer et al., 2018), which provides atlases at multiple resolutions (100–1000 parcels).

In related work, Park et al. (2023a) evaluated glioma-reference differences in rsFC using conventional graph-theoretic measures (eg, modularity, clustering coefficient, degree centrality) (Bullmore and Sporns, 2009; Rubinov and Sporns, 2010) computed across Schaefer parcellations from 100 to 1000 parcels. Park et al. (2023a) and colleagues showed that patient-reference dissimilarity becomes increasingly apparent at finer parcellation granularity then plateaued around 400 parcels, highlighting the strong dependence of standard graph metrics on parcellation reso-

lution. This raises a fundamental question: do patient-control differences reflect a *scale-specific* phenomenon, or is there an underlying signal that manifests similarly across resolutions?

1.4 Statistical Connectomics

Conventional analysis of brain networks relies on summary statistics—modularity, clustering coefficient, degree centrality, path length—that reduce a connectome to a handful of scalar values (Bullmore and Sporns, 2009; Rubinov and Sporns, 2010). Most rs-fMRI studies implementing graph-theoretic methods use these standard heuristic metrics to characterize connectivity differences (Rubinov and Sporns, 2010; Alexander-Bloch et al., 2012).

In glioma specifically, Park et al. (2023a) provide a systematic summary of rs-fMRI studies using standard graph-theoretic analyses, finding that all identified studies rely on such conventional measures. While computationally convenient, this approach has fundamental limitations. First, networks with *wildly different topologies* can produce identical summary statistics (Chen et al., 2019)—observing reduced modularity in patients does not specify *how* the network is disrupted. Second, as Park et al. (2023a) demonstrated, conventional graph metrics show strong dependence on parcellation resolution, making it unclear whether observed effects reflect biology or atlas choice.

Statistical connectomics provides an alternative: replace heuristic measures from combinatorial graph theory with model-based statistics from random graph theory (Chung et al., 2021; Gopalakrishnan et al., 2024). This shift—from descriptive measures to model-based estimation via eigendecomposition—yields methods with theoretical guarantees (consistency, asymptotic normality) that conventional approaches lack (Athreya et al., 2018; Rubin-Delanchy et al., 2022).

The random dot product graph (RDPG) model (Young and Scheinerman, 2007; Athreya et al., 2018) provides the mathematical foundation. Under this model, each node possesses a latent position in a low-dimensional Euclidean space, with edge probabilities determined by inner products of these positions. The omnibus embedding framework (Levin et al., 2017) extends spectral methods to simultaneously embed multiple networks into a shared coordinate system, enabling principled comparison across subjects.

Spectral embedding preserves the full multivariate geometry of connectomes. Two networks with identical conventional graph statistics will occupy different positions in the embedding space if their edge-level structure differs; the Frobenius distance between embeddings captures topological variation that summary statistics miss. Applying multidimensional scaling (MDS) to these distances produces a “map of networks”—a low-dimensional visualization where each connectome is a single point, and proximity reflects similarity in the complete FC architecture (Athreya et al., 2022; MacDonald et al., 2024).

1.5 Present Work

We apply omnibus spectral embedding to FC data from GBM patients and healthy reference subjects across six parcellation resolutions. Our analysis reveals three main findings:

1. The patient-reference separation is **scale-invariant**: the same subjects are identified as “patient-like” or “reference-like” regardless of parcellation resolution, with cross-scale correlations exceeding $r = 0.95$.
2. A continuous **patient-likeness score** derived from the spectral embedding correlates more strongly with FC edges than binary group labels, demonstrating that the embedding captures graded FC variation beyond categorical diagnosis.
3. The FC signature of patient-likeness is **interpretable**: reduced somatomotor and visual network connectivity characterizes high-scoring individuals in *both* patient and reference groups.

Table 1: Cohort demographics and clinical characteristics.

Characteristic	GBM Patients ($n = 217$)	Controls ($n = 252$)
Age, mean $\pm$ SD (years)	[TBD]	[TBD]
Sex (% male)	[TBD]	[TBD]
Overall survival, median (days)	425	—
IDH1 wildtype (%)	94.5%	—
MGMT methylated (%)	41.2%	—

2 Methods

2.1 Study Cohorts

GBM patients ($n = 217$) were recruited from the neurosurgery brain tumor service at Washington University Medical Center. All patients had newly diagnosed, histopathologically confirmed glioblastoma (WHO grade IV) and completed preoperative resting-state fMRI. Age-matched healthy controls ($n = 252$) were drawn from the OASIS-3 cohort (LaMontagne et al., 2019). This patient population has been characterized in prior studies (Park et al., 2023b). Table 1 summarizes cohort characteristics.

2.2 Image Acquisition and Preprocessing

Resting-state fMRI data were preprocessed using previously described pipeline (Park et al., 2023a). Briefly, this included slice timing correction, correction of odd-even slice intensity differences, and rigid-body motion correction within and across runs. Preprocessed rs-fMRI data were then resampled into 3mm isotropic atlas space in register with each subject’s T1-weighted image via a combination of an initial affine transform and a non-linear ANTs warp with cost-function masking to exclude tumor tissue, as detailed in (Park et al., 2023a). Additional steps included voxel-wise detrending, temporal low-pass filtering (0.15Hz), regression of nuisance signals (6 head motion parameters, CSF, and whole-brain signal), and spatial smoothing with a 6mm full-width-half-maximum Gaussian kernel. Subjects with excessive motion were excluded. [Additional preprocessing details to be added by K.Y.P.]

2.3 Parcellation

We employ the Schaefer parcellation (Schaefer et al., 2018) at six granularities: 100, 200, 300, 400, 500, and 600 parcels, organized according to the 7-network scheme of Yeo et al. (2011): Visual (VIS), Somatomotor (SMN), Dorsal Attention (DAN), Ventral Attention/Salience (SAL), Limbic (LIMB), Frontoparietal Control (CON), and Default Mode (DMN). We exclude limbic network parcels due to susceptibility artifacts (Ojemann et al., 1997), and therefore all subsequent analyses focus on the remaining six networks (VIS, SMN, DAN, SAL, CON, DMN), yielding 95, 188, 280, 374, 467, and 558 parcels at each granularity, respectively.

2.4 Functional Connectivity Matrices

For each subject, we extract the mean BOLD time series for each parcel and compute the Pearson correlation matrix $\mathbf{F} \in \mathbb{R}^{K \times K}$:

$$\mathbf{F}_{ij} = \frac{\sum_{t=1}^T (\bar{\mathbf{B}}_{it} - \mu_i)(\bar{\mathbf{B}}_{jt} - \mu_j)}{\sqrt{\sum_{t=1}^T (\bar{\mathbf{B}}_{it} - \mu_i)^2} \sqrt{\sum_{t=1}^T (\bar{\mathbf{B}}_{jt} - \mu_j)^2}} \quad (1)$$

where $\mu_k = \frac{1}{T} \sum_{t=1}^T \bar{\mathbf{B}}_{kt}$ is the temporal mean for parcel k . This yields $m = 469$ FC matrices $\{\mathbf{F}^{(1)}, \dots, \mathbf{F}^{(m)}\}$ at each parcellation resolution.

2.5 Statistical Network Analysis Pipeline

Our analysis pipeline proceeds in four stages: (1) omnibus spectral embedding, (2) Frobenius distance computation, (3) multidimensional scaling, and (4) projection onto the discriminant axis. We describe each stage below.

2.5.1 Random Dot Product Graph Model

We adopt the random dot product graph (RDPG) framework (Young and Scheinerman, 2007; Athreya et al., 2018). Under this model, each vertex i is associated with a latent position $\mathbf{X}_i \in \mathbb{R}^d$, and edge probabilities (or expected weights) are determined by inner products:

$$\mathbb{E}[\mathbf{A}_{ij}] = \mathbf{X}_i^\top \mathbf{X}_j \quad (2)$$

The RDPG is analytically tractable while approximating a wide range of network structures, including the stochastic block model (Alyakin et al., 2024).

2.5.2 Adjacency Spectral Embedding

The key inferential tool for RDPGs is adjacency spectral embedding (ASE), which estimates latent positions via eigendecomposition (Sussman et al., 2012). In this analysis, for each subject, the rs-FC matrix $\mathbf{FC} \in \mathbb{R}^{K \times K}$ (K = number of parcels) is treated as a symmetric weighted adjacency matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ with $n = K$. Given $\mathbf{A}$ with eigendecomposition $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$, the d -dimensional ASE is:

$$\hat{\mathbf{X}} = \mathbf{U}_d |\mathbf{\Lambda}_d|^{1/2} \quad (3)$$

where $\mathbf{U}_d$ contains the d eigenvectors corresponding to the d largest-magnitude eigenvalues. Under the RDPG model, ASE provides consistent estimates of the latent positions up to orthogonal transformation (Sussman et al., 2012).

2.5.3 Omnibus Embedding

Comparing spectral embeddings across subjects is complicated by rotational non-identifiability: any orthogonal transformation of the latent positions preserves inner products. The omnibus embedding (Levin et al., 2017) addresses this by jointly embedding multiple networks.

Given m adjacency matrices $\mathbf{A}^{(1)}, \dots, \mathbf{A}^{(m)} \in \mathbb{R}^{n \times n}$ on a common vertex set, the omnibus matrix $\mathbf{M} \in \mathbb{R}^{mn \times mn}$ is defined as:

$$\mathbf{M}_{ij} = \begin{cases} \mathbf{A}^{(i)} & \text{if } i = j \\ \frac{1}{2}(\mathbf{A}^{(i)} + \mathbf{A}^{(j)}) & \text{if } i \neq j \end{cases} \quad (4)$$

Applying ASE to $\mathbf{M}$ yields aligned embeddings $\hat{\mathbf{X}}^{(1)}, \dots, \hat{\mathbf{X}}^{(m)} \in \mathbb{R}^{n \times d}$ that reside in a common coordinate system.

We select embedding dimension d using the automatic profile likelihood method (Zhu and Ghodsi, 2006), which identifies the elbow in the scree plot. Across all parcellation resolutions, this procedure selected $d = 8$ dimensions.

2.5.4 Frobenius Distance Between Connectomes

The aligned embeddings enable a natural distance metric between networks. For subjects i and j , we compute the Frobenius norm:

$$D_{ij} = \|\hat{\mathbf{X}}^{(i)} - \hat{\mathbf{X}}^{(j)}\|_F = \sqrt{\sum_{k=1}^n \sum_{\ell=1}^d (\hat{X}_{k\ell}^{(i)} - \hat{X}_{k\ell}^{(j)})^2} \quad (5)$$

This yields an $m \times m$ pairwise distance matrix $\mathbf{D}$, which we term the “network of networks.”

2.5.5 Multidimensional Scaling

Classical multidimensional scaling (MDS) applied to $\mathbf{D}$ produces a low-dimensional representation where each point corresponds to an entire connectome. We embed into 2 dimensions for visualization and subsequent analysis.

2.5.6 Patient-Likeness Projection

To obtain a continuous measure of FC disruption, we fit a linear support vector machine (SVM) to the 2D MDS coordinates:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_i \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i \quad (6)$$

where $y_i \in \{-1, +1\}$ indicates group membership. The signed distance from each subject to the decision boundary defines the *patient-likeness score*:

$$p_i = \frac{\mathbf{w}^\top \mathbf{x}_i + b}{\|\mathbf{w}\|} \quad (7)$$

Subjects with $p_i < 0$ fall on the control side; those with $p_i > 0$ fall on the patient side. Crucially, p_i is continuous: it captures *how far* each subject lies from the boundary, not merely which side.

2.6 Correlational Analyses

2.6.1 Edge-Level FC Correlations

To characterize which FC connections relate to patient-likeness, we leveraged the continuous patient-likeness scores p_i (Methods 2.5.6). For each FC entry across the $K(K-1)/2$ unique parcel pairs, we computed the Pearson correlation between that edge’s FC value across all 469 subjects and the corresponding patient-likeness scores. For comparison, we also correlate FC edges with the binary group label (0 = control, 1 = patient). To assess statistical significance, we randomly permuted group-labels or patient-likeness scores 1000 times to generate null distributions of correlations, and empirical p-values were adjusted for multiple comparisons using FDR correction.

2.6.2 Network-Level Aggregation

We aggregate edge-level correlations by network pair. For each pair of networks (e.g., SMN_L–VIS_R), we compute the mean correlation across all edges connecting parcels in those networks.

2.6.3 Extremes Analysis

To visualize what “patient-likeness” means in FC terms, we compare mean FC matrices for subjects at the extremes of the projection distribution: the top and bottom 15% within each group (controls and patients separately).

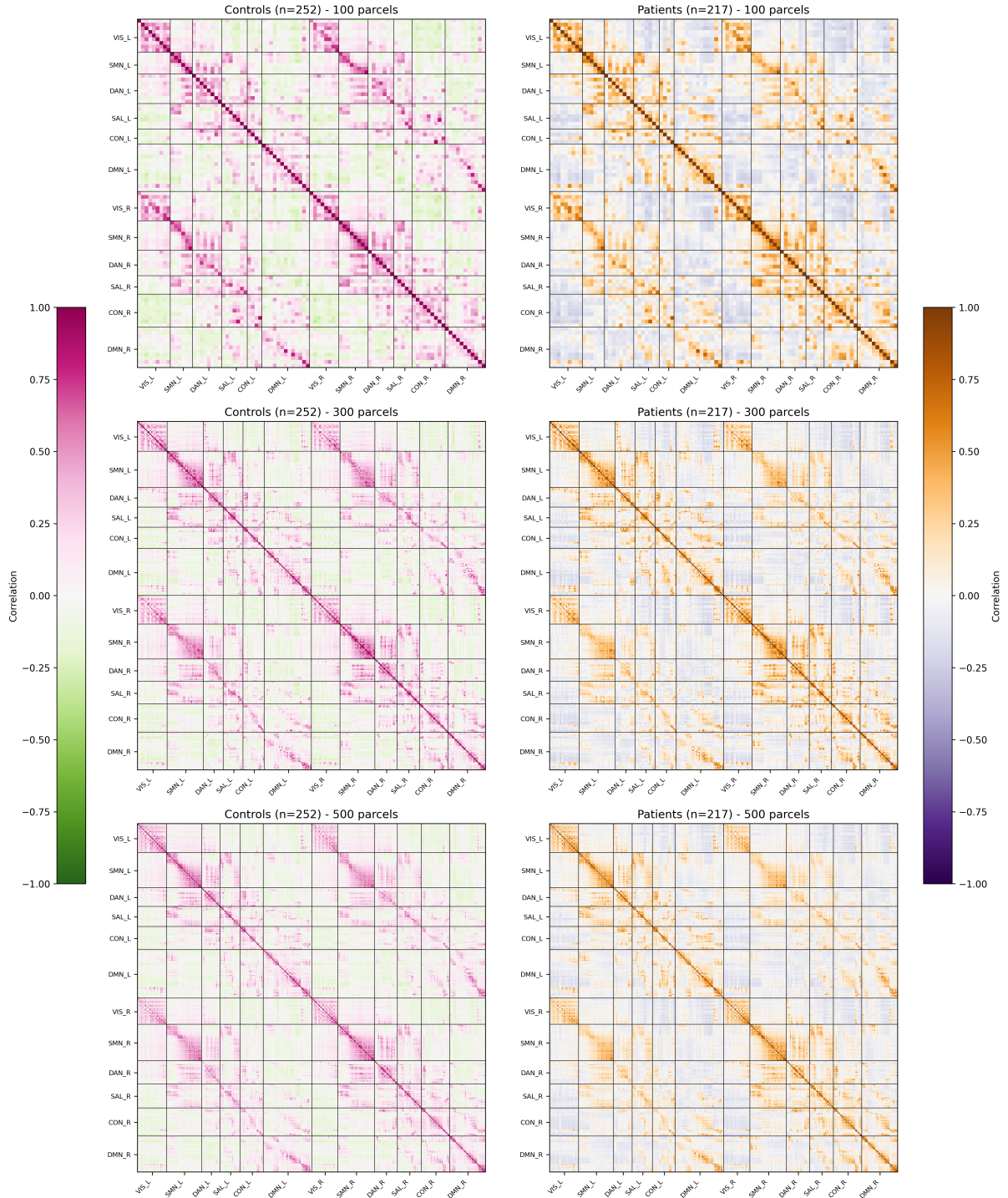


Figure 1: **Mean functional connectivity matrices for controls and patients across parcellation resolutions.** Each heatmap displays the group-averaged Pearson correlation between all parcel pairs at 100 parcels (top row), 300 parcels (middle row), and 500 parcels (bottom row). Left column: controls ($n = 252$); right column: patients ($n = 217$). Rows and columns are ordered by resting-state network assignment: VIS = Visual, SMN = Somatomotor, DAN = Dorsal Attention, SAL = Salience, CON = Control, DMN = Default Mode; L/R indicate left/right hemisphere. Diagonal blocks correspond to within-network connectivity; off-diagonal blocks correspond to between-network connectivity.

3 Results

3.1 Functional Connectivity Structure

Figure 1 displays mean FC matrices for controls and patients at three representative parcellation resolutions. The canonical block-diagonal structure is evident in both groups, reflecting the organization of the Schaefer parcellation into functional networks. Visually, the matrices are strikingly similar—the characteristic pattern of strong within-network connectivity (diagonal blocks) and weaker between-network connectivity is preserved in patients. Figure 2 compares mean parcel embeddings between patients and reference subjects. The distributions overlap substantially—consistent with the visual similarity of the FC matrices.

Figure 3 shows the mean parcel-level embedding at 100 parcels, colored by network assignment. The first latent dimension roughly separates lower-order networks (VIS, SMN, DAN) from higher-order association networks (SAL, CON, DMN), indicating that the omnibus embedding recovers a gradient from sensory/attentional to control/default systems in the latent space.

3.2 Group Separation in the Spectral Embedding

Figure 4 displays the 2D MDS representation of connectome distances at each parcellation resolution. Patients and controls form overlapping but distinguishable distributions. Linear SVM classification achieves 72–75% accuracy (5-fold cross-validation) across all resolutions, with Cohen’s d ranging from 1.09 to 1.24 (Table 2).

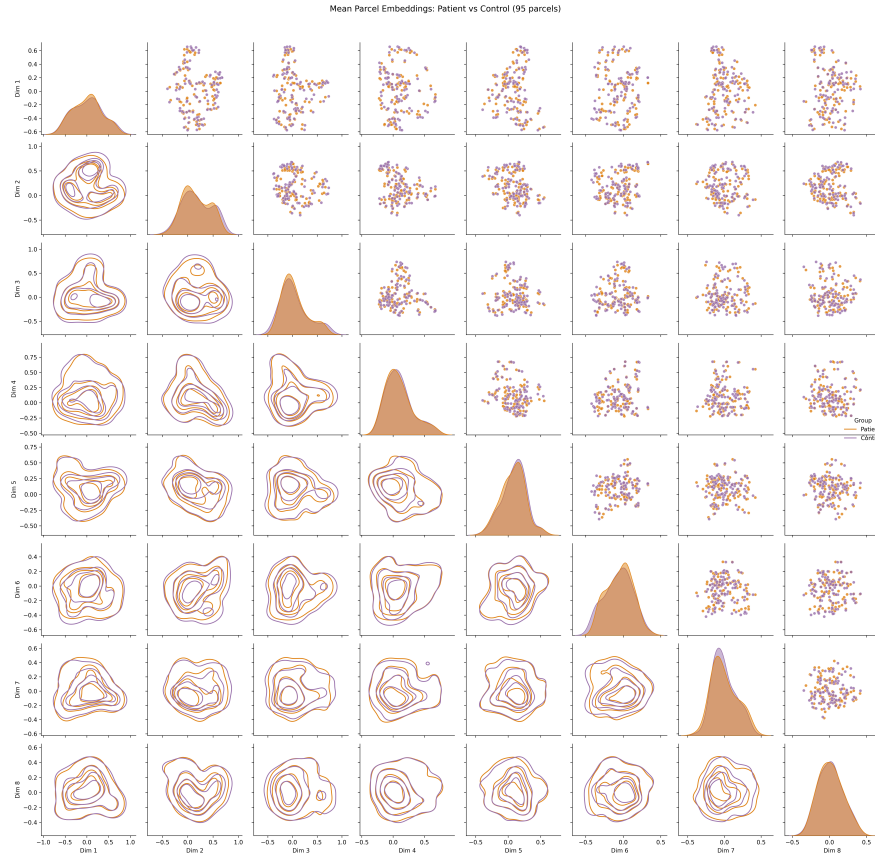


Figure 2: **Pairplot comparing mean parcel omnibus latent-space embeddings between patients and controls.** Orange: mean patients; purple: mean controls. 95 parcels after excluding limbic network. Diagonal: kernel density estimates of parcel positions along each dimension. Upper triangle: scatter plots of parcel positions. Lower triangle: contour plots of the joint densities. The distributions largely overlap.

3.3 Scale-Invariance of Patient-Likeness

The patient-likeness score is highly consistent across parcellation granularities (all cross-scale correlations $r = 0.88$; mean $r = 0.95$; Figure 5). This stability is accompanied by an identical selected embedding dimension ($d=8$) across six parcellation (Table 2). These findings suggest that the omnibus embedding captures a similar low-dimensional latent structure across scales and that patient-reference separation in this latent space is relatively robust to parcellation granularity.

Effect sizes are similarly stable: Cohen’s d ranges from 1.09 to 1.24 across parcellation granularities (coefficient of variation: 4.1%). If anything, effect size *decreases* slightly at finer granularities, opposite to what would be expected if fine-grained details drove the separation.

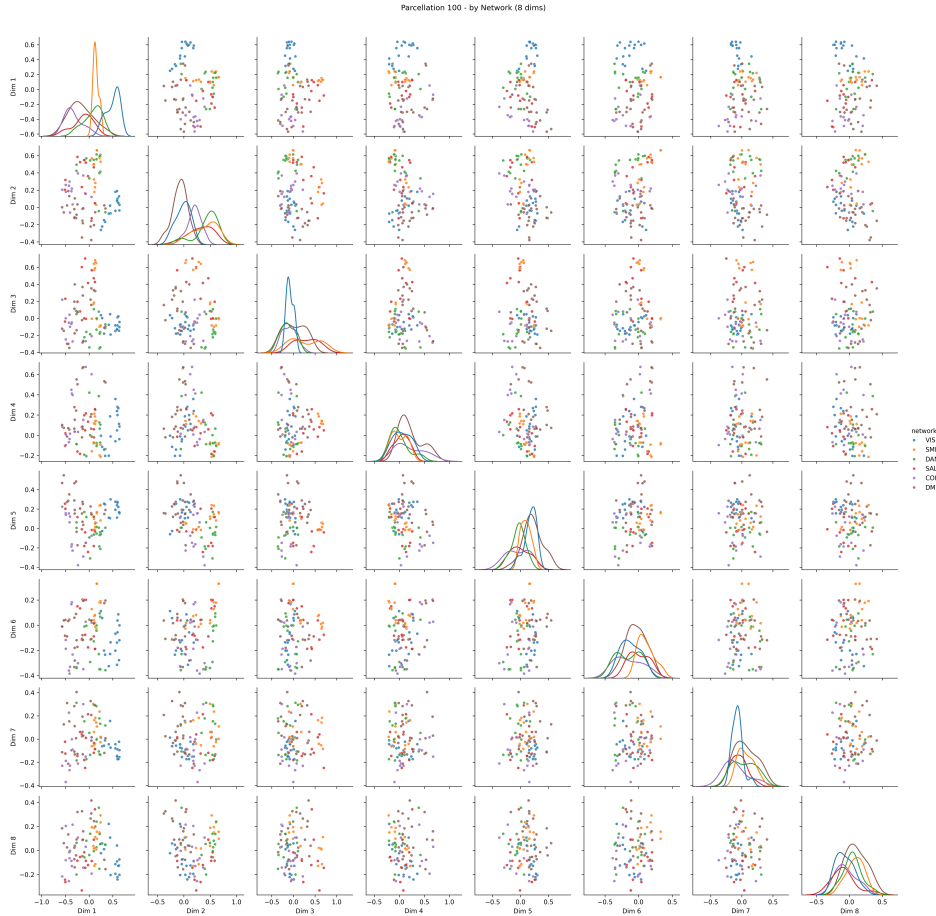


Figure 3: Omnibus latent-space embedding preserves resting-state network structure. Each point represents a single brain parcel from the 100-parcel Schaefer atlas, embedded in a shared latent space using omnibus adjacency spectral embedding and then averaged across all subjects. Parcels are colored by their Yeo 6-network assignment (excluding Limbic). Diagonal panels show how parcels are distributed along each embedding dimension. Off-diagonal panels show that parcels from the same resting-state network cluster together and occupy distinct regions of the latent space. For example, in the first-row panels, visual (VIS) parcels cluster in a distinct band and are separated from default mode (DMN) parcels.

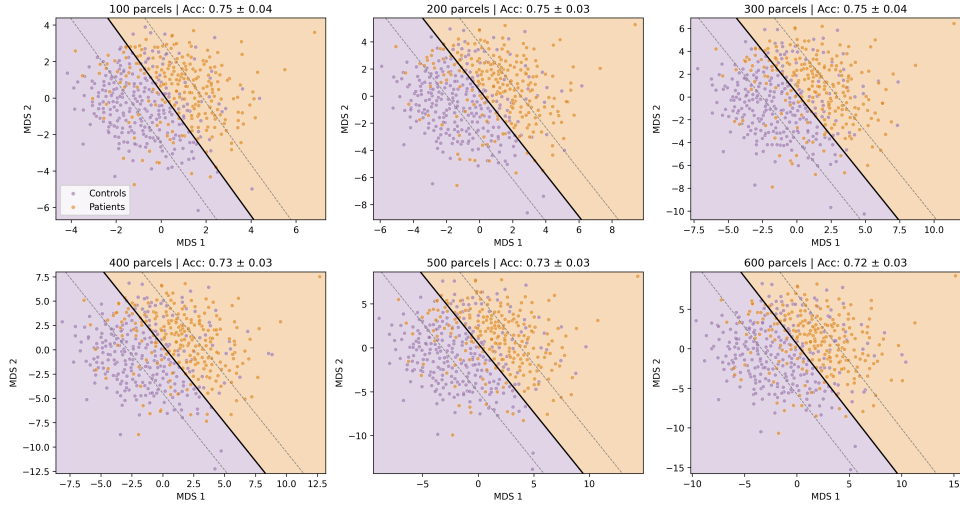


Figure 4: **Two-dimensional MDS representation of connectome distances with linear SVM decision boundaries.** Each panel corresponds to a different parcellation resolution: 100 parcels (top-left), 200 parcels (top-middle), 300 parcels (top-right), 400 parcels (bottom-left), 500 parcels (bottom-middle), and 600 parcels (bottom-right). Each point represents one subject's connectome projected into 2D via multidimensional scaling of Frobenius distances between omnibus embeddings. Purple: controls ($n = 252$); orange: patients ($n = 217$). The solid diagonal line shows the linear SVM decision boundary; dashed lines indicate the SVM margin.

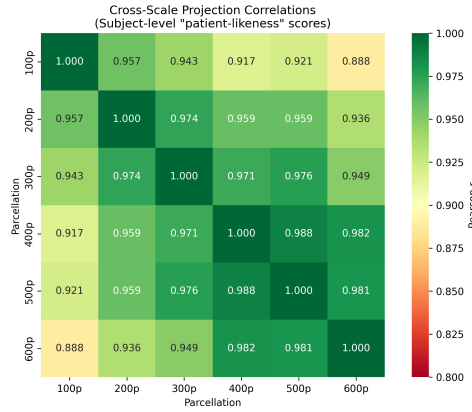


Figure 5: **Cross-scale correlation matrix of subject-level patient-likeness scores.** Each cell displays the Pearson correlation (r) between patient-likeness projection scores computed at two different parcellation resolutions.

Table 2: Effect sizes and classification performance across parcellation resolutions.

Parcellation	100	200	300	400	500	600
Omni Embedding dim.	8	8	8	8	8	8
Cohen’s d	1.24	1.23	1.19	1.17	1.15	1.09
CV Accuracy (%)	75	75	75	73	73	72

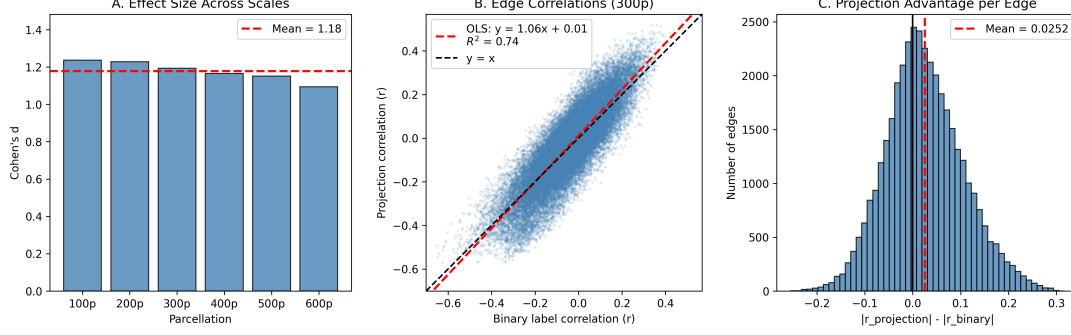


Figure 6: (A) Cohen’s d across parcellation resolutions. (B) Scatter plot of edge-level correlations: projection score vs. binary label (300 parcels). Points above the diagonal indicate stronger correlation with the projection. Wald’s test ($H_0 : \beta = 1$) $p < 10^{-80}$ (C) Distribution of the per-edge advantage ($|r_{\text{proj}}| - |r_{\text{binary}}|$). The positive mean indicates systematic advantage for the projection score.

3.4 Patient-Likeness Captures More Than Group Membership

If the patient-likeness score merely reflected binary group membership, correlations with FC edges should equal those obtained using the group label directly. Instead, the projection score correlates *more strongly* with FC. At 300 parcels (39,060 edges), the mean absolute correlation is $|\bar{r}| = 0.146$ for the projection score versus $|\bar{r}| = 0.121$ for binary labels—a 21% increase. The projection achieves higher $|r|$ for 60.8% of edges (Wilcoxon signed-rank test: $p < 10^{-10}$). Figure 6B–C illustrates this advantage. Importantly, the two measures are strongly correlated ($r = 0.86$ across edges), indicating they capture related signal. The projection *sharpens* rather than replaces the group difference: edges that distinguish patients from controls also correlate with the continuous score, but with larger effect sizes.

Table 3 summarizes the number of FDR-significant edges across parcellations. The projection identifies 10–20% more significant connections than binary comparison at each resolution.

Table 3: Significant FC edges (FDR $q < 0.05$) by analysis method.

Parcellation	100	200	300	400	500	600
Total edges	4,465	17,578	39,060	69,751	108,811	155,403
Projection (sig.)	2,718	10,548	22,548	38,442	60,064	84,049
Binary (sig.)	2,309	8,980	19,218	32,668	50,946	69,497
Prop. increase	17.7%	17.5%	17.3%	17.7%	17.9%	20.9%

3.5 Patient-Likeness Reflects a Continuous FC Phenotype

The patient-likeness score exhibits substantial within-group variance. For example, at 300 parcels, 11.1% of controls score above the patient mean, and 11.1% of patients score below the control mean.

These “misclassified” individuals are not errors—they represent the tails of a continuous distribution. To test whether “patient-like” reference subjects genuinely resemble patients in their FC architecture, we compared mean FC matrices for subjects at the extremes of the projection (Figure 7).

The difference maps (right column of Figure 7) reveal the FC signature of patient-likeness. High-scoring individuals—whether patients or reference subjects—exhibit:

- Reduced connectivity within and between somatomotor (SMN) and visual (VIS) networks
- Reduced inter-hemispheric SMN–VIS networks

Crucially, the difference pattern is consistent across groups: the within-controls difference correlates at $r = 0.63$ with the within-patients difference ($p < 10^{-10}$). The same edges define “patient-likeness” regardless of actual diagnosis.

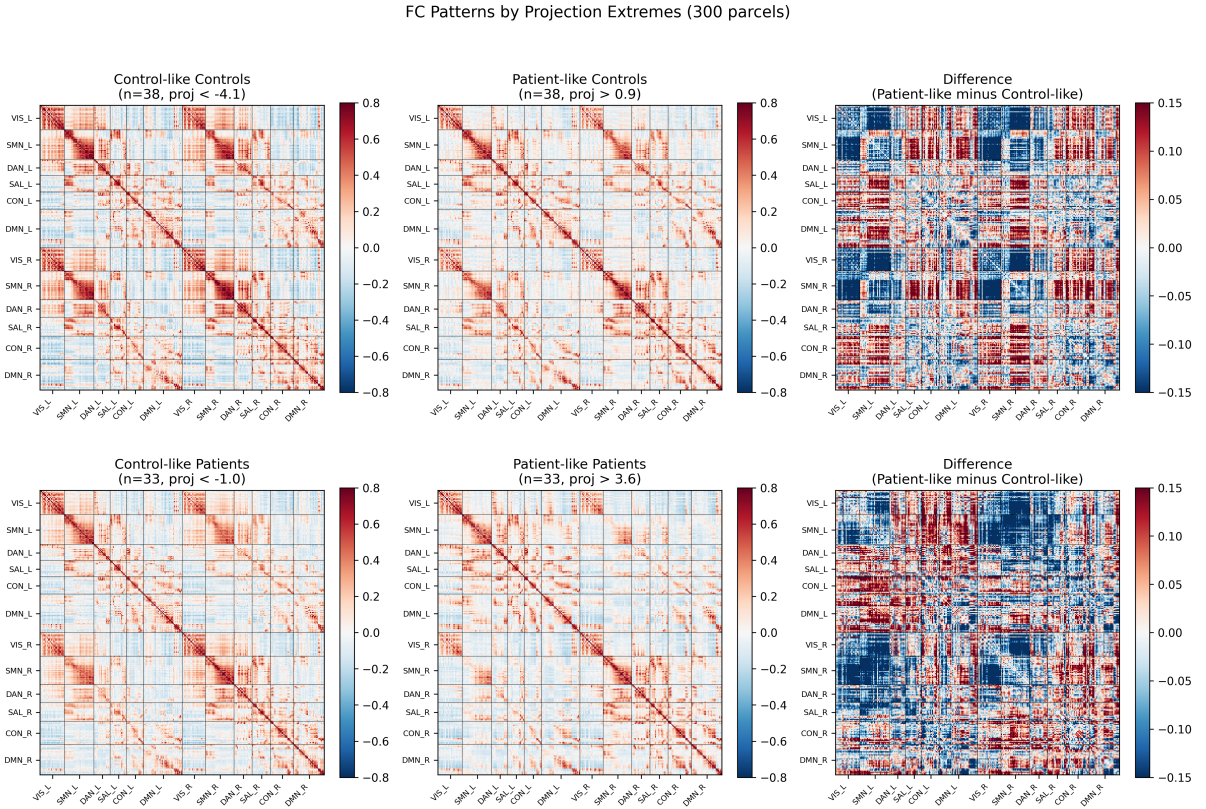


Figure 7: Mean FC matrices for subjects at extremes of the patient-likeness distribution (300 parcels, top/bottom 15% within each group). Left: subjects with low projection scores (“control-like”). Middle: subjects with high projection scores (“patient-like”). Right: difference (patient-like minus control-like). Top row: within controls only. Bottom row: within patients only. The difference patterns are strikingly similar across groups.

3.6 Network-Level Characterization

Figure 8 and Table 4 present network-level effects. The dominant pattern is reduced connectivity within and between VIS, SMN, and to a lesser extent DAN, indicating broadly decreased sensorimotor-visual coupling. Conversely, CON with VIS/SMN show modest increases in patient-like individuals, suggesting relatively preserved or enhanced control-sensorimotor coupling.

Within-hemisphere effects are weaker but follow the same pattern: SMN and VIS connections are reduced, while connections involving frontoparietal control (CON) networks show the opposite—modest *increases* in patient-like individuals. This dissociation suggests that patient-likeness is not simply global connectivity loss, but a specific reorganization: sensorimotor integration is reduced while control network coupling is relatively preserved or enhanced.

The bilateral symmetry is notable given that GBM is a unilateral disease. Tumors in the left hemisphere do not preferentially affect left-sided networks. This supports the interpretation that patient-likeness reflects a distributed, whole-brain phenotype rather than focal disruption near the lesion.

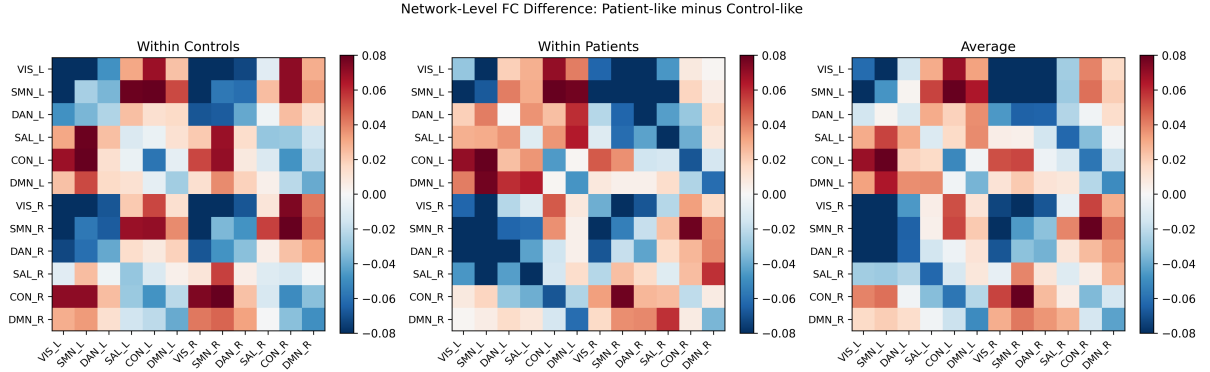


Figure 8: Network-level FC differences at 300 parcels. Left: within controls. Middle: within patients. Right: average. Blue indicates reduced connectivity in patient-like individuals; red indicates increased connectivity. The pattern is consistent across groups.

Table 4: Network-level FC differences (patient-like minus control-like, averaged across groups). Negative values indicate reduced connectivity in patient-like individuals.

Most Decreased		Most Increased	
Network Pair	Δr	Network Pair	Δr
VIS_L-SMN_R	-0.170	SMN_L-CON_L	+0.098
VIS_R-SMN_R	-0.169	SMN_R-CON_R	+0.083
SMN_L-VIS_R	-0.164	VIS_L-CON_L	+0.069
VIS_L-SMN_L	-0.126	SMN_L-DMN_L	+0.065
SMN_L-SMN_R	-0.114	VIS_R-CON_R	+0.054

4 Discussion

4.1 Summary of Findings

We applied statistical connectomics methods to compare FC between patients with glioblastoma and reference subjects. Three main findings warrant discussion:

Scale-invariance. Subject-level patient-likeness scores correlate at $r > 0.95$ across parcellation resolutions from 100 to 600 parcels. The omnibus embedding consistently recovers an 8-dimensional latent structure, suggesting the patient-reference separation reflects a low-dimensional signal intrinsic to the FC data rather than an artifact of parcellation granularity.

Continuous phenotype. The projection onto the patient-reference discriminant correlates more strongly with FC edges than binary diagnostic labels. The 21% increase in mean $|r|$ indicates the embedding extracts graded FC variation beyond categorical group membership. Substantial within-group variance (11% of each group falls on the “wrong” side of the mean) supports interpreting the projection as a continuous phenotype rather than a noisy classifier.

Interpretable signature. Patient-likeness is characterized by reduced somatomotor and visual network connectivity, particularly in inter-hemispheric connections. The same FC pattern defines patient-likeness in both patients and controls ($r = 0.63$ between within-group difference maps), suggesting the projection captures a phenotype that exists as normal variation and is exaggerated in disease.

4.2 Relation to Prior Work

Our network-level findings align with prior work on GBM functional connectivity. Luckett et al. (2024) found that somatomotor, visual, and auditory network connectivity predicted functional outcomes (Karnofsky Performance Status) in high-grade glioma patients, achieving 94% classification accuracy. Daniel et al. (2021b) showed that homotopic connectivity in SMN and DAN was associated with overall survival, with lower SMN and DAN homotopic connectivity predicting shorter survival. These results support the idea that alterations in extrinsic networks—including somatomotor, visual, auditory, and dorsal attention systems—are clinically relevant and align with the patient-likeness phenotype we report.

The scale-invariance finding highlights a limitation of conventional graph-theoretic approaches. Standard graph theoretic metrics summarizing FC have been shown to be highly sensitive to parcellation choice ((Park et al., 2023a)), making it difficult to know whether observed group differences reflect true FC differences or arbitrary granularity. In contrast, the omnibus spectral embedding captures a stable 8 dimensional latent structure across Schaefer parcellations, and the patient-reference separation is preserved with increasing parcellation granularity. This suggests that the group difference is carried by a low-dimensional FC phenotype that traditional graph metrics are not designed to capture.

From a methodological perspective, our work builds on the statistical connectomics framework of Chung et al. (2021) and the multiscale analysis of Gopalakrishnan et al. (2024). We extend these approaches by introducing the patient-likeness projection as a tool for interpreting what drives group separation in the spectral embedding.

4.3 Why Somatomotor and Visual Networks?

treat with caution until head motion artifacts and etc.

The prominence of SMN and VIS in our results warrants discussion. Several factors may contribute:

Network properties. Somatomotor and visual networks are spatially compact, have strong within-network connectivity, and exhibit robust inter-hemispheric coupling. These properties may make them sensitive to distributed disruption.

Clinical presentation. Motor and visual deficits are common in GBM patients, particularly those with tumors near eloquent cortex. FC alterations in these networks may reflect both direct tumor effects and compensatory reorganization.

Signal quality. Sensorimotor regions have high signal-to-noise ratio in fMRI due to proximity to the head coil and distance from susceptibility artifacts. This could increase sensitivity to detect genuine effects.

Notably, higher-order networks (default mode, frontoparietal control) showed weaker or positive associations with patient-likeness. This may reflect the hierarchical organization of brain networks: unimodal sensory-motor networks are more tightly constrained by anatomy, while transmodal networks have greater capacity for functional reorganization.

4.4 Limitations

Several limitations should be acknowledged. The RDPG model assumes low-rank structure that may not fully capture the complexity of brain networks. We did not model tumor location, which could influence which networks are most affected. The patient cohort comes from a single institution; multi-site validation is needed. Finally, we cannot establish causality: the FC phenotype could reflect premorbid variation, disease effects, or their interaction.

++ - no explicit head motion control yet (send it to anton) - fyi anton, increased head motion in the scanner disproportionately affect vis/smn/dan - the analyses relating fc matrices/patient-likeness x adjust for age, sex, tumor vol, tumor location - so adding a simple analysis whether pt-like FC reflects these factors need to be looked at (you have the excel sheet) - FC computed including parcels/voxels including tumor so pt matrices not a clean sample really - signal from lesional tissue/peri-tumoral distortion could be driving part of the observed pattern

4.5 Future Directions

The continuous patient-likeness score could serve as a biomarker in several contexts. In surgical planning, patients with “control-like” FC might have better preserved functional architecture. In clinical trials, the projection could serve as a continuous endpoint more sensitive than binary outcomes. The existence of “patient-like” controls raises questions about whether this phenotype predicts risk or resilience.

5 Conclusion

Statistical connectomics reveals a scale-invariant, low-dimensional FC phenotype that separates GBM patients from healthy controls. The spectral embedding extracts richer signal than mass-univariate analysis, captured by a continuous patient-likeness score that correlates strongly with somatomotor and visual network connectivity. The same FC signature defines patient-likeness in both populations, suggesting the projection captures a dimension of normal variation that is exaggerated in disease. These findings support the conceptualization of glioblastoma as a network-level disease and demonstrate the utility of spectral methods for discovering interpretable biomarkers of distributed FC disruption.

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Code Availability

Analysis code utilizes the `graspologic` library (Chung et al., 2019) and is available at [repository].

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